

Challenge (Habanero)

- Q1.** A gamer's high score s in points after t hours of playing is modeled by $s(t) = -2(t - 5)^2 + 100$. What is the vertex form of this function?
- (A) $s(t) = -2(t - 5)^2 + 100$
(B) $s(t) = -2t^2 + 20t + 0$
(C) $s(t) = 2(t - 5)^2 - 100$
(D) $s(t) = 2t^2 - 20t - 100$
(E) $s(t) = -2(t + 5)^2 - 100$
- Q2.** The data usage d in GB of a gaming app after x hours is given by $d(x) = x^2 - 4x - 5$. What is the intercept form of this function?
- (A) $d(x) = (x - 5)(x + 1)$
(B) $d(x) = (x + 5)(x - 1)$
(C) $d(x) = (x - 1)(x + 5)$
(D) $d(x) = x^2 - 4x - 5$
(E) $d(x) = (x - 2)(x - 2)$
- Q3.** A quadratic function is in the form $f(x) = a(x - h)^2 + k$, where (h, k) is the vertex. If $f(x) = -3(x - 2)^2 + 10$, what is the value of k ?
- (A) -3
(B) 2
(C) 10
(D) -10
(E) 3
- Q4.** The graph of $y = x^2 - 6x + 5$ has x -intercepts at $x = 1$ and $x = 5$. What is the intercept form of this function?
- (A) $y = (x - 1)(x + 5)$
(B) $y = (x - 1)(x - 5)$
(C) $y = x^2 - 6x + 5$
(D) $y = (x - 5)(x - 1)$
(E) $y = (x + 1)(x + 5)$
- Q5.** The score s of a player in an online game is modeled by $s(t) = -t^2 + 4t + 2$. What is the vertex of this function?
- (A) $(2, 6)$
(B) $(1, 3)$
(C) $(0, 2)$
(D) $(4, 2)$
(E) $(-2, 6)$
- Q6.** A gaming app's battery usage b in percent after t hours is given by $b(t) = (t - 2)(t + 1)$. What is the vertex form of this function?
- (A) $b(t) = (t - 2)(t + 1)$
(B) $b(t) = t^2 - t - 2$
(C) $b(t) = (t + 1)(t - 2)$
(D) $b(t) = -t^2 + t + 2$
(E) $b(t) = -(t - 0.5)^2 + 2.25$
- Q7.** A quadratic function $f(x) = ax^2 + bx + c$ has x -intercepts at $x = -2$ and $x = 4$. What is the intercept form of this function?
- (A) $f(x) = (x + 2)(x + 4)$
(B) $f(x) = a(x + 2)(x - 4)$
(C) $f(x) = a(x - 2)(x + 4)$
(D) $f(x) = (x - 2)(x + 4)$
(E) $f(x) = (x + 2)(x - 4)$
- Q8.** The function $f(x) = -2(x - 3)^2 + 5$ represents the score of a player. What is the value of a in this function?
- (A) -2
(B) 2
(C) 3
(D) 5
(E) -5

Open Ended Questions (FRQ)

- Q9.** Write a quadratic function $f(x)$ in vertex form to model the data usage of a gaming app. Assume the vertex is $(2, 10)$ and the function opens downwards. Provide the equation of the function and explain the meaning of each parameter.
- Q10.** The gaming app's score is modeled by the function $s(t) = (t - 1)(t + 3)$. Rewrite the function in vertex form and find the vertex. Explain the meaning of the vertex in this context.

Chef's Tips

- Emphasize the importance of identifying the vertex and intercepts in quadratic functions.
- Use real-world examples from tech and gaming to illustrate the applications of vertex and intercept forms.
- Encourage students to use graphing calculators or software to visualize the functions and verify their answers.